JASON LU

Research engineer: post-training, inference optimization, and RL environment & eval infrastructure for self-improving AI

New York, NY | +1 917-615-8837 | hl4072@columbia.edu | github.com/HaochengLu | haochenglu.github.io

EDUCATION

Columbia University | *M.S. in Computer Science* — *New York, NY* | Expected Dec. 2027
NYU Shanghai | *B.S. in Computer Science* | *GPA 3.84 / 4.00* — *Shanghai, China* | Jun. 2026

EXPERIENCE

Evolvent AI
May 2026 – Present
Research Assistant, Recursive Self-Improvement Infrastructure — *Shanghai, China*

- **Agentic data synthesis engine**: Architected a production-scale, benchmark-agnostic engine that turns diverse benchmarks and real-world engineering issues into verified agentic post-training data via multi-agent orchestration and closed-loop verification; powers recurring commercial deliveries to **ByteDance, Alibaba (Qwen), and Zhipu AI (GLM)**.
- **RSIGym** [paper · code]: Co-led a budgeted environment where frontier agents run AI research end to end, from data synthesis to LoRA training and eval. Co-designed **RSI-Index** (mean share of remaining gap closed, 5 benchmarks): best agent **0.4809** at $500/benchmark, taking a 35B base model's **SWE-bench Verified 17.67%** → **50.33%** and AIME 31.67% → 97.78%.
- **RSIBench-Data** [paper · code]: Co-designed a frozen-budget benchmark of agents as post-training data researchers (35B MoE): across **4 agents** × **6 benchmarks**, **SWE-bench Verified 12%** → **46%**, yet **18 of 23** searches past peak ended lower.
- **Gödel Forest** [paper · code]: Co-built multi-agent RSI in which parallel search trees co-evolve via ∼100-token shared lessons; **+10.70 pp** over a single agent across 6 benchmarks (**SWE-bench Verified 35%** → **54%**).

Intent (GroupUltra)
Feb. 2026 – Aug. 2026
Agent Systems Engineering Intern, personal & workplace agents — *Remote*

- Shipped Intent's multiplayer team agent ahead of Anthropic's Claude Tag, with the same core loop: anyone in a group chat @-mentions one shared agent that keeps channel-wide memory, uses tools, runs long tasks async, and follows up proactively.
- Scaled personal and workplace agents with persistent persona, memory, and relationship graphs to **10K+ users** in 1K+ groups: **100+ agents**, 1M+ interactions, **80%+** autonomous decisions.
- Designed the restart-safe durable runtime behind **24h+** agent tasks (checkpoints, leases + watchdogs, idempotency keys + side-effect fencing): **99%+** failure recovery over 100K+ jobs / 1M+ tool calls, **<0.01%** duplicate critical effects.
- Gated every release on deterministic replay of production traces: core-task success **+30%**, manual intervention **−60%**.

DP Technology
Mar. 2025 – Feb. 2026
Algorithm Engineering Intern, Uni-Parser team — *Shanghai, China*

- **Uni-Parser** [paper · site]: Core contributor to an industrial multi-expert engine that turns scientific papers and patents into LLM and AI4Science training corpora; built its pipeline-parallel inference path (micro-batching, async GPU dispatch, load balancing) and scaled it to **240 L40 GPUs**, parsing **16M+ documents in 6 days** with near-linear scaling up to 1,000 GPUs.
- **MolParser-Mobile** [paper · model]: Co-designed an AutoML-searched **9.98M-param** OCSR model whose cross-attention decoder reuses cached visual K/V: **1,520 mol/s on one GPU**, **38**× faster than MolParser 1.0 at higher BioVista accuracy.
- Ran the full training recipe: pretraining on **18M+** image→E-SMILES pairs, SFT with replay, on-policy distillation from MolParser 1.5, and DPO on chirality/Markush errors (pattern-match **52%** → **78%**).

RESEARCH PROJECTS

CrashGym | *Python, NumPy, scikit-learn*
NeurIPS 2026 Agenthon Workshop · [paper]

- Built a deterministic order-book crash simulator (price impact, liquidity withdrawal, margin liquidation, evolving strategy mix) where LLM agents experiment on a lab copy before choosing an intervention, scored by simulator-computed regret over every feasible action; lab access cut regret **57.4%** on 24 prospective worlds.

Continual-Learning SWE Benchmark | *Python, Docker, coding-agent eval*

- Rebuilt **500** SWE-bench/SWE-rebench tasks into dependency curricula: each keeps its real snapshot and tests but gains an injected issue the original fix can't solve, needing a rule or persisted state from an earlier task.
- Each task ships a new full repair and private tests, gated by executable checks (original fix passes the old task, fails the new one; full repair passes both); frontier agents' solve rate fell from **57% to 24%**.

Verifier-Guided Hard-Negative DPO | *PyTorch, TRL, QLoRA*
[paper]

- Built MathVerifier, a 6-dim CoT error profiler (step alignment, NLI, SymPy; no LLM judge), to distill ∼600K traces into 10K weighted hard-negative DPO pairs; beat random-pair DPO by **3.0 pts** on GSM8K and MATH (**80.1% / 67.1%**).

ContextVC | *Rust, MCP, agent hooks*
140+ GitHub stars

- Rust CLI that makes coding-agent memory code: compiles one Git-versioned `.context/` source into 6 agents' native rule files, gates tool calls via `PreToolUse` hooks/MCP, and fails CI on drift.

SKILLS

Training & inference: PyTorch, Hugging Face/TRL, LoRA/QLoRA SFT, DPO, on-policy distillation; GPU batching & scheduling
Infra & languages: Python, Rust, C/C++, TypeScript, SQL; Harbor, E2B, Docker, Linux, MCP, CI/CD, durable job runtimes